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CERTIFICATE

This is to certify that the dissertation entitled

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In partial fulfilment for the completion of Fifth-semester Capstone Project Phase 1 (UE24AM320A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering (AI & ML) under the rules and regulations of PES University, Bengaluru during the period August - December 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the fifth-semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase 1 entitled “Securing Stigmergic Communication in Cooperative Multi-Agent Reinforcement Learning Swarms” has been carried out by us under the guidance of BHASKARJYOTI DAS, Project Guide, and submitted in partial fulfilment of the course requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI & ML) of PES University, Bengaluru during the academic semester August - December 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

Cooperative agents can communicate indirectly through shared environmental traces. This approach, called stigmergy, avoids direct messages but also allows compromised agents to add misleading signals. We study this problem in a simulated food-retrieval grid with a parameter-shared PPO policy and strictly local observations. The proposed defense direction uses local pheromone history, revisitation and own task progress; no learned defense is claimed at this stage.

Phase 1 establishes the simulation, fair clean baselines, frozen-policy attack evaluation and a data pipeline. Candidate J, a 12 by 12 grid with four agents and 100 food units, passed the clean stochastic development gate against separately trained no-pheromone PPO and random actions for three training seeds. A fresh run reproduced these means. The original three bounded attack settings did not establish average damage. Stronger sustained injection reduced mean honest delivery from 70.33 to 59.81 units, about 15%, but only 18 of 36 matched pairs lost deliveries, so the declared consistency gate still failed. A separately controlled decoy route also failed to establish mean injection damage.

A recorded development corpus contains 108 matched episodes and an export of 412,016 agent-step rows. Its local observations and twelve history features are separated from provenance, transition outcomes and privileged audit metadata. The newer 432 diagnostic episodes remain separate. Exposure labels, final grouped partitions, temporal windows, detector training and held-out evaluation are future work. The present contribution is a tested evaluation and data foundation with explicit evidence boundaries, rather than a completed robustness solution.

Keywords: stigmergy; cooperative reinforcement learning; PPO; pheromone injection; local observations; trajectory corpus

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CHAPTER 1

INTRODUCTION

1.1 Project context

A team of agents can coordinate by changing a shared environment. Ant trails are a familiar example: an ant leaves a signal, and another ant reacts to that signal later. This form of indirect communication is called stigmergy. In our project, the environment is a simulated grid, and the signals are numerical food and home pheromone fields. Agents search for food, carry one unit at a time and return it to a nest.

A shared field is useful because agents do not need to send direct messages. It also creates a security problem. A compromised agent can leave a signal that looks like an ordinary trail even when it does not correspond to food. Earlier research already demonstrated misleading pheromone attacks and a cautionary-pheromone response [1]. Our project therefore does not claim that false pheromone or minority disruption is a new idea.

Our question concerns a learned team rather than a prescribed ant controller. Research on learned stigmergy shows that environmental traces can support collaboration, although its tasks and learning arrangements differ from ours [2]. We use a parameter-shared Proximal Policy Optimization policy, or PPO [3]. Each agent applies the same network to its own local view. There is no global-state critic, direct communication channel or attacker identity in its input.

1.2 Purpose of Phase 1

Phase 1 establishes a research question, a critical literature comparison, a working simulation and a reproducible evaluation and data pipeline. It also tests two prerequisites: whether pheromones improve clean retrieval, and whether the bounded attack causes enough disruption to support a meaningful defense study. A working injector alone cannot answer the second question.

The implementation work is divided into Phase 2A and Phase 2B. These are internal work-package names within academic Capstone Project Phase 1; they are not the university's Phase I and Phase II. Phase 2A checks clean pheromone benefit using separately trained baselines. Phase 2B freezes the selected policies, measures attack effects and prepares development trajectories for inspection.

1.3 Current position

Candidate J passed the declared clean stochastic development gate for three training seeds. A fresh run on 6 November 2026 reproduced the earlier clean means. Stronger sustained injection subsequently produced a mean honest-delivery loss of 10.53 food units, but only 18 of 36 matched pairs lost deliveries. Every declared attack-consistency gate still failed. These findings justify further investigation, not a claim of reliable disruption or successful mitigation.

The corpus builder and agent-step CSV exporter are implemented. The historical corpus is development evidence, not a final labelled detector dataset. Exposure labels, final grouped partitions, temporal windows, learned detector, learned defense and held-out evaluation remain pending. The measured results and their limits are reported in Chapter 7.

CHAPTER 2

PROBLEM DEFINITION

2.1 Problem statement

We study whether an agent can use only its local pheromone history, revisitation and own task progress to reduce disruption from bounded false-food deposits while preserving clean cooperative retrieval. The immediate experimental problem is to establish a useful learned clean policy and measure injection damage without confusing it with changes in worker movement, seed selection or field mechanics.

Agents observe a 3 by 3 neighbourhood rather than the complete grid. Food and home pheromone concentrations do not identify the agent that created them. A high concentration can result from a genuine productive trail, overlapping honest trails, waiting behaviour by carrying agents, or malicious deposits. Concentration alone is therefore not a confirmed attack label.

2.2 Threat and information boundaries

The primary attacker controls bounded, nonnegative food-pheromone deposits at its actual occupied cell. Per-step and episode budgets, field clipping and evaporation constrain the applied signal. It cannot erase trails, create negative concentration, write at arbitrary global coordinates, change food or rewards, or modify the honest policy's weights.

The original persistent profiles retain the selected agent's productive PPO movement. A separately declared local decoy experiment changes only that selected agent's movement. Its disabled and attacked controls share the same movement rule. This distinction matters because diverting a worker can reduce delivery even if its false pheromone has no additional effect.

Runtime defense inputs must remain local. Simulator knowledge of attacker selection, injection events, global food and total delivery is reserved for audit and future target construction. It cannot become a policy or detector feature. The current implementation has a local timeout heuristic, but no fitted detector or learned mitigation policy.

2.3 Objectives and success criteria

The clean-policy objective is a strict stochastic mean-delivery advantage over both a separately trained no-pheromone PPO and a random-action reference for every declared training seed. The attack objective is separately evaluated: positive overall mean honest loss, positive mean for every

training seed and a majority of matched pairs showing positive loss. These are development gates, not statistical significance tests.

The final defense study must compare undefended coordination, separately trained no-pheromone PPO, the local timeout baseline and a documented cautionary-pheromone adaptation. It must measure clean-performance cost as well as attack recovery. Those complete research comparisons are not yet available.

2.4 Scope and possible applications

The implemented scope is simulated two-dimensional cooperative resource retrieval. Warehouse collection, exploration with limited communication and distributed inspection are potential application domains. In each, unreliable shared cues could waste movement or repeated visits. These are motivations for the study, not deployments. Physical robots would require different sensing, motion and safety models and independent validation.

CHAPTER 3

LITERATURE SURVEY

3.1 Survey scope and organisation

This report makes a focused critical comparison of eight core research papers. Papers are grouped by their role in the problem: pheromone attacks, learned coordination, field and sensing limits, adversarial multi-agent learning, and the PPO method. Software documentation and repository links are listed separately and are not counted as papers. The faculty's broader target of more than thirty relevant papers remains incomplete.

The comparison asks what each work establishes, which assumptions differ from our setting and which experimental choice follows for our project. This is more useful than treating every citation as evidence that our specific policy must coordinate successfully or that its attack must work.

3.2 Pheromone attacks and mitigation

Aswale et al. study misleading pheromone in simulated foraging and introduce cautionary pheromone to limit disruption [1]. This is the closest attack and mitigation precedent. It establishes that environmental signals can be manipulated by a minority and that an additional local signal can help under the studied controller. It does not establish the performance of our parameter-shared PPO. We must measure our clean policy, injection effect and mitigation cost directly.

3.3 Learned stigmergic coordination

Xu et al. combine digital pheromone with independent learning, conflict avoidance and a federated training arrangement for collaborative shape formation [2]. This supports learned environmental communication, but its task and controller are not our resource-retrieval PPO. It motivates a separately trained no-pheromone comparison instead of assuming that adding a field automatically helps.

3.4 Field mechanics and sensing limits

Hunt, Jones and Hauert investigate pheromone-based exploration as swarm density increases [4]. Their findings show that the value of stigmergy depends on operating conditions. For our study, field saturation and repeated local visits need to be measured rather than automatically treated as malicious behaviour.

Phormica demonstrates an artificial pheromone release and sensing system using photochromic material and robot swarms [5]. It provides physical context for environmental signals. Our project uses numerical fields and does not reproduce that hardware or claim its physical sensing results. Its supplementary material is related evidence rather than a ready-made dataset for our frozen PPO attack protocol.

3.5 Further coordination and security methods

Aina and Ha study learned coordination using virtual pheromones and curriculum learning in a congested retrieval setting [6]. Their independent Q-learning and curriculum differ from our shared PPO. The work is relevant to task design and coordination limits; it does not establish our bounded attack's strength or a local defense result.

Standen, Kim and Szabo organise execution-time attacks and defenses in multi-agent reinforcement learning [7]. Their taxonomy motivates explicit statements of attacker access, timing and defense information. A survey supports the evaluation framework; it is not a source of labelled pheromone trajectories for this project.

Li et al. demonstrate adversarial minority influence through a learned action-based attack [8]. The attacker exploits interactions among cooperative agents. This capability differs from our primary bounded field injector. Its success cannot be transferred to our experiment by assumption. We retain separate controls when adding a movement capability.

The PPO paper describes policy optimisation using sampled interactions and a clipped update objective [3]. It supplies the learning method, not evidence of swarm-security robustness. Our implementation uses Stable-Baselines3 with one shared actor and a local value network; it does not implement a centralised global critic.

3.6 Findings across the eight papers

Table 3.1 connects the literature to the project decisions. The remaining question is a candidate evaluation gap, not a claim that every component is novel or that no relevant public data exists.

Table 3.1 Area-wise literature findings

Area and core papers	What is established	Consequence for our study
Pheromone attacks [1]	Misleading signals and cautionary mitigation	Do not claim injection or cautionary pheromone as new

Area and core papers	What is established	Consequence for our study
Learned coordination [2,6]	Environmental traces can support learning	Verify benefit under our PPO and task
Mechanics and sensing [4,5]	Density and sensing assumptions matter	Measure local signal visibility and benign ambiguity
Security and PPO [3,7,8]	Learning methods and attack access differ	Keep controls, capability and evidence boundaries explicit

CHAPTER 4

RESEARCH AND TECHNOLOGY GAPS AND CHALLENGES

4.1 Candidate gap

The reviewed sources do not establish the complete evaluation proposed here: a parameter-shared PPO using local pheromone observations, bounded occupied-cell injection, leakage-separated matched trajectories and a defense based only on local history and own progress, tested under unseen conditions. The candidate contribution is a reproducible evaluation and data protocol for that combination. Its research value depends on results and the broader literature survey.

A public resource may cover swarm movement, adversarial MARL or paper metadata without providing our exact trajectories. We therefore state a qualified dataset finding. We have not proved that a matching dataset is absent everywhere, and dataset absence alone would not be a contribution.

4.2 Main technical challenges

First, a clean policy may retrieve food without making useful use of pheromone. Hiding channels only after training can also produce an unfamiliar input distribution. Our full and no-pheromone PPO models instead learn separately from the start with otherwise matched settings.

Second, visible injection is not the same as harmful injection. An attacker retaining productive movement may reinforce a useful route. Stronger injection can change actions without reducing delivery in every case. Local field and policy-response diagnostics help distinguish these possibilities without treating them as detector scores.

Third, attack-control comparisons can mix different causes. If the attacker stops working, clean-minus-attacked delivery includes worker diversion. The new decoy variant therefore uses a disabled-decoy twin and reports worker cost separately.

Fourth, correlated trajectories create leakage risks. The same map, policy and action seed appear in related scenarios, and temporal windows may overlap. Final groups and map partitions must be assigned before windows, not by randomly shuffling rows after feature generation.

4.3 Questions still open

Reliable attack degradation, validated lingering-exposure targets, clean false-alarm rates and local defense recovery remain unproven. The mixed deterministic policy results are another limitation. Future work must use an accepted validation protocol rather than repeatedly increasing budgets or selecting maps until a favourable gap appears.

CHAPTER 5

REQUIREMENT SPECIFICATION

5.1 Functional requirements

The simulator must support seeded maps, legal local actions, pickup and delivery, food conservation, shared reward and bounded food/home fields. Every live agent must receive a local observation of the documented shape. Training must use one shared PPO policy and save a reloadable checkpoint with the actual transition count.

Evaluation must freeze validated checkpoints, run complete matched scenario families and report honest and team delivery separately. The injector must log requested, authorised and applied mass without revealing attacker metadata in local inputs. Corpus creation and CSV export must reject incomplete or unverifiable records instead of treating them as completed data.

Table 5.1 Functional and reproducibility requirements

Requirement	**Implemented mechanism**	**Evidence or limit**
Local policy input	58-value per-agent observation	No global map, identity or direct message
Fair clean comparison	Full and masked models trained separately	Same maps, rewards and PPO settings
Matched attack measurement	Clean, disabled and attacked families	Worker routing controlled explicitly
Data separation	Local records and privileged audit files	CSV exporter does not open audit steps
Reproducibility	Seeds, hashes, immutable output directories	Partial and failed status retained
Final defense evaluation	Grouped splits and required baselines	Full protocol remains future work

5.2 Non-functional requirements

Runs must be reproducible on the declared dependency stack, while cross-platform byte identity is not promised. Configuration changes and invalid completed-run records must not silently pass resume checks. Observations, history features and rewards must be finite; actions and Boolean outcomes must satisfy their contracts. Complete families, unique group IDs and consistent map/seed provenance are mandatory.

Every substantial code file has an explanatory header and important functions have responsibility and parameter documentation. Decision records precede changes to mechanics, attack capability or evaluation. Negative and interrupted results remain available. Recorded test counts are associated with their executed dates rather than presented as fresh checks during document preparation.

5.3 Hardware and software requirements

The verified training stack uses Python 3.12, NumPy, Gymnasium, PettingZoo [9], Stable-Baselines3 2.7.1 [10] and PyTorch 2.5.1 [11]. Pytest supports regression testing; JSON/JSONL and CSV support recording and inspection; matplotlib supports figures; Git and GitHub support source history. CPU training is supported. A GPU is optional and no special hardware or physical robot is required for the implemented scope.

The repository has pinned dependency files. A working Python 3.12 virtual environment should be created from those files before running the suite or training commands. Local environments can contain machine-specific interpreter paths and should not be assumed portable. Installation and small smoke checks should precede expensive experiments.

CHAPTER 6

PROPOSED SYSTEM ARCHITECTURE

6.1 System block diagram

The architecture has a clean-training stage, a frozen-policy evaluation stage and a data stage. Figure 6.1 shows the current flow and the planned local detector boundary. The audit branch is for provenance and future target construction; it does not supply runtime observations.

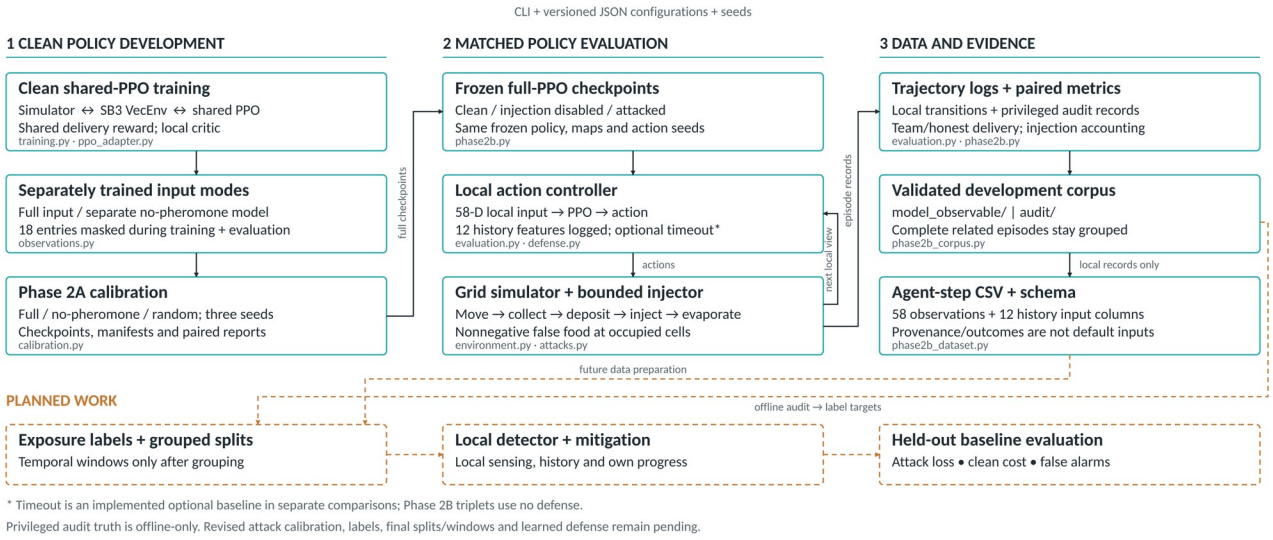


Figure 6.1 Training evaluation and data flow

6.2 Major modules

The environment module implements the grid transition and creates local observations. The PPO adapter presents one observation row per agent to Stable-Baselines3. These rows belong to agents in a coupled world; they are not four independent environments. The training module saves configuration, source hashes, initial and final checkpoints, progress and diagnostic summaries.

The calibration runner compares full input, separately trained no-pheromone input and random actions. The attack/evaluation modules load completed frozen checkpoints and run matched conditions. The primary injector writes only at selected agents' actual post-movement cells, with the same clipping and evaporation as ordinary fields.

The corpus builder validates complete scenario families and packages local and audit streams separately. The dataset exporter reads local files and emits an agent-step CSV plus a schema. LocalHistoryFeatures uses only current observations and the agent's own preceding actions. The

existing timeout component is a rule-based baseline; the planned detector and learned defense are not implemented.

6.3 Step ordering and field equations

One environment step validates the entire action batch, moves agents, handles pickup/delivery in seeded contention order, deposits ordinary pheromone, applies permitted false-food injection, clips both fields and evaporates them. It then returns the next observation and own progress events. A carrying agent arriving at the nest delivers before the deposit channel is chosen.

For field concentration P at cell x, ordinary deposit D and false-food injection I, the update is:

$$P(t+1,x) = (1-e)\min(C, P(t,x) + D(t,x) + I(t,x)) \quad (6.1)$$

Here e is the evaporation fraction and C is the field cap. I is zero for the home field and for runs with injection disabled. Concentration is a dimensionless simulation quantity, not a measured chemical concentration. Candidate J uses e = 0.02 and C = 10.

A carrying agent deposits d = 1 into the food field at its occupied cell, including when stationary. An empty agent inside the nest deposits the ordinary home amount. Outside the nest, an empty agent deposits home pheromone only after a successful move:

$$D_{home} = d \times 0.95^{k} \text{ for a successful outside move; } 0 \text{ for outside waiting} \quad (6.2)$$

The counter k records successful outside movement since the last nest visit and resets at the nest. This rule anchors deposits to nest visits but does not guarantee a perfect global gradient after trails overlap. Food deposition is unchanged by this home rule.

6.4 Local observation and decision flow

Each 3 by 3 cell patch contains six channels: food presence, nest presence, normalised food field, normalised home field, normalised occupancy and boundary. Nine cells times six channels gives 54 values. Carrying, pickup, delivery and blocked flags give 58 in total. Actions 0 through 4 are stay, north, east, south and west. No absolute location or trail-author identity is supplied.

At time t, the agent receives O_t and selects A_t. The world returns $O_{(t+1)}$, reward and outcomes. History H_t must use only information available by t. Using the reward or pickup caused by A_t as an input to the same decision would leak future information. Logged outcomes remain separate from default candidate detector inputs.

CHAPTER 7

IMPLEMENTATION

7.1 Implementation overview

Phase 1 now includes a working simulator, shared PPO training, clean calibration, bounded attack evaluation, local diagnostics, corpus packaging and CSV export. Later audit corrections strengthened paired-configuration validation, resume integrity and local/provenance contracts. These changes preserve mechanics, rewards and policy settings rather than changing them to improve an apparent score.

Table 7.1 Implementation modules and outputs

Module	Responsibility	Main output
environment.py and attacks.py	Grid transition and bounded injection	Local observations and audit accounting
ppo_adapter.py and training.py	Shared PPO learning and reload	Checkpoints, progress and manifests
observations.py and calibration.py	Input masking and fair comparisons	Full/no/random results and paired differences
phase2b.py and attack_diagnostics.py	Frozen evaluation and signal checks	Matched losses, action response and mass records
phase2b_corpus.py	Validate and package complete families	Separate local and audit trees
phase2b_dataset.py and defense.py	Export local agent rows and history	CSV, schema and local features

7.2 Dataset description and strategy

Candidate external resources include benign stigmergy experiments, artificial-pheromone supplementary records and broader adversarial MARL benchmarks. They inform design but do not directly supply trajectories matching our policy, observation contract and bounded occupied-cell

attacker. We selected simulator-generated matched episodes as the data-generation strategy. No final detector-training dataset has been finalised.

The previously recorded development corpus contains three historical calibrations: immediate budget 20, immediate budget 200 and budget 200 delayed until transition 201. Each has three policies times four maps times one action seed, giving twelve matched groups. Each group has clean, injection-disabled and attacked episodes. Across all settings this gives 36 groups and 108 episodes. Reused maps mean these are not 36 unique maps or independent final-test samples.

The recorded export contains 103,004 world transitions. Four agents contribute one row each per transition, producing 412,016 agent-step rows. Episodes can end early, so the transition count is not 108 times the maximum horizon. These historical corpus counts are retained from the recorded build; the full original corpus was not regenerated during ESA preparation.

Table 7.2 Agent-step CSV column families

Column family	**Count**	**Intended use**
Local observation	58	Current local input
Local-history features	12	Candidate detector input; not extra PPO channels
Provenance	10	Grouping, tracing and audit; excluded by default
Transition outcomes	9	Actions, reward and resulting events; separate from inputs
Total	89	Development data view; no exposure targets yet

The twelve history features are food/home field mean and maximum, change in both means, carrying, pickup, delivery, blocked, normalised time without progress and revisit fraction. Simulator attacker identities, injection events/mass and global delivery remain in audit files. Scenario IDs and seeds are provenance, not valid detector predictors. The 432 newer diagnostic episodes use a separate experimental format and have not automatically been merged into the historical corpus.

The next dataset steps are validated residual-exposure targets, grouped map/family partitions, temporal windows and only then detector fitting. A deposit event alone does not describe exposure after the injected field lingers or is clipped. Final sample counts, split percentages and window sizes

have not been declared. Learned-defense trajectories can be collected only after an actual detector/mitigation mechanism is evaluated.

7.3 Clean-policy methodology and results

Candidate J uses a 12 by 12 grid, four agents, a top-left 2 by 2 nest and two distant adjacent food cells with fifty units each. The horizon is 1,100 steps, evaporation is 2% per step, the ordinary deposit is 1 and the cap is 10. The adjacent cluster location changes with the map seed.

Full PPO receives all 58 inputs. A separate no-pheromone PPO trains from scratch with the same eighteen food/home pheromone entries zeroed during both training and evaluation. Other local channels remain available. The same reward, two 32-unit hidden layers for actor/local critic, maps and optimiser settings apply. Training seeds are 7, 17 and 27; training maps are 0–15 and diagnostic maps are 100–103. Each model requests 300,000 agent transitions and records 300,032 after rollout rounding. Random actions provide a non-learned reference.

Table 7.3 Candidate J stochastic mean team deliveries

Training seed	Full PPO	No-pheromone PPO	Random	Full minus no-pheromone
7	100.00	68.75	3.00	+31.25
17	80.00	63.50	1.75	+16.50
27	100.00	80.50	2.25	+19.50

Each mean covers four diagnostic maps. All three full policies exceed both references, satisfying the declared clean stochastic gate. Fresh training on 6 November reproduced the prior means exactly. Deterministic full-policy means remain 100, 0 and 60, so the result does not establish reliable deterministic coordination or generalisation.

Earlier candidates A–I are retained in decision records. They gave mixed full/no-pheromone results, a 60-unit capacity tie or unstable shorter-horizon learning. Candidate J retained the stable long horizon and increased only food capacity from thirty to fifty per patch relative to Candidate G. This tested a specific ceiling explanation while leaving the learner unchanged.

7.4 Attack calibration and follow-up results

For productive-movement settings, clean and injection-disabled runs have identical local behaviour; only attacked runs inject false-food mass. Honest-delivery loss is defined as:

L_injection = honest delivery(disabled) - honest delivery(attacked) (7.1)

A positive value is damage; a negative value means the attacked episode delivered more. It does not prove attacks are generally helpful or harmless. Team delivery includes all four agents, while honest delivery excludes the same selected agent in both controls. One selected attacker out of four is 25%, distinct from the historical eight-agent Review 1 fixture.

The 6 November follow-up first recreated the missing clean checkpoints and froze them. It repeated the three historical profiles and added two predeclared sustained profiles. A separate decision permitted a local east-edge decoy movement rule, with worker diversion measured separately. All 144 matched groups and 432 evaluation episodes completed within the declared experiment budget. No new training or attack matrix was run for this ESA report.

Table 7.4 All completed attack settings and honest losses

Profile	**Request per step and episode cap**	**Mean loss**	**Positive pairs**
Immediate historical	1; cap 20; start immediately	-2.17	1 of 12
Longer historical	1; cap 200; start immediately	-0.42	2 of 12
Delayed historical	1; cap 200; transition 201	-2.25	1 of 12
Sustained one	1; cap 1,100; start immediately	+2.47	13 of 36
Sustained three	3; cap 3,300; start immediately	+10.53	18 of 36
Local east-edge decoy	3; cap 3,300; changed routing	-1.47	9 of 36

The historical profiles use action seed 7; the new profiles use action seeds 7, 8 and 9. All use the three policy seeds and four development maps. On the common action-seed-7 subset, sustained-three loss is +10.08 against -0.42 for the historical immediate-200 setting. The expanded means are not identical samples and should not be presented as such.

Under sustained-three, honest delivery falls from 70.33 to 59.81, about 15.0%, and mean team loss is 17.44. Per-policy-seed honest losses are +8.92, +22.42 and +0.25. However, eighteen positive

pairs are exactly half, not a majority; six pairs tie and twelve improve. The preregistered consistency gate therefore still fails. The larger dose is explicitly different from the old budget, and no statistical significance or held-out result is claimed.

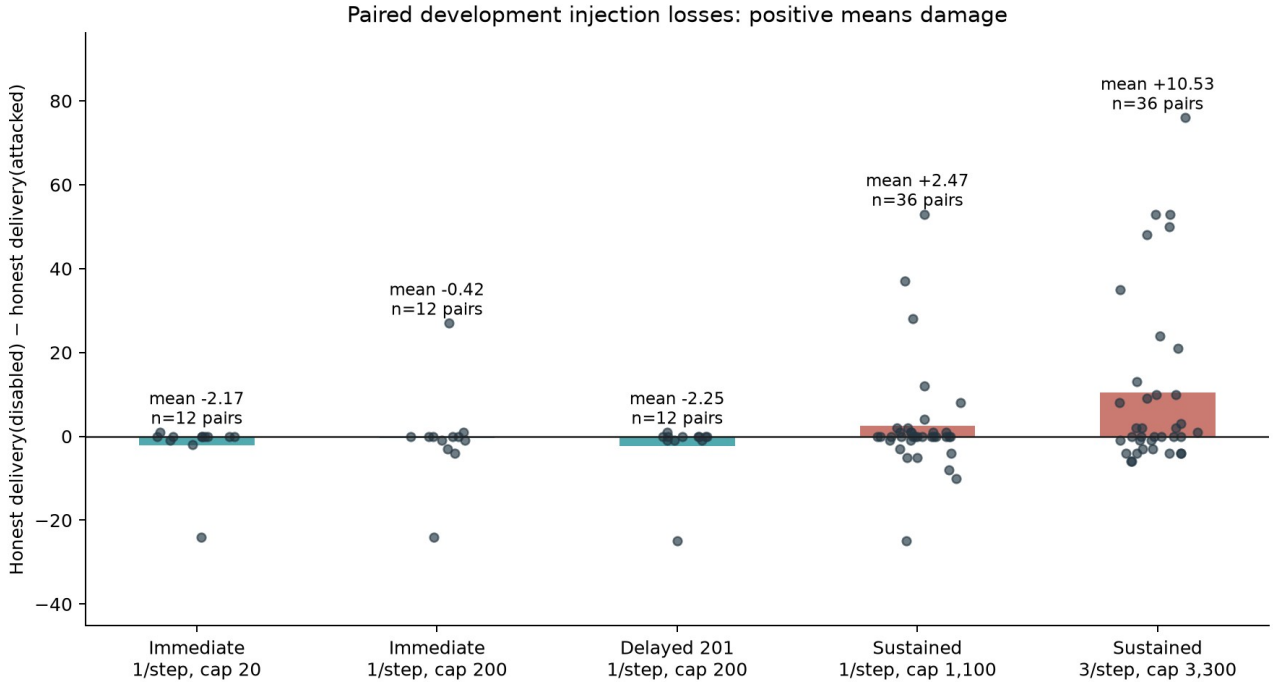


Figure 7.1 Every paired persistent-injection loss

The local decoy agent proposes PPO actions but executes east unless its own east-neighbour boundary bit requires staying. Honest actions remain unchanged PPO, and no global food target is used. Both disabled and attacked twins have the same decoy route. Worker diversion costs 15.03 mean team deliveries, while injection loss is -1.47. This variant did not establish harmful pheromone injection. Comparing clean directly with attacked would incorrectly mix worker and injection effects.

7.5 What the diagnostics measure

An audit-only shadow field replays the actual positions and ordinary food deposits with the same clipping/evaporation but without injection. At each pre-action observation, the actual local food crop is compared with that shadow. A second policy forward pass measures action-distribution response when only the nine food-pheromone entries are replaced. It does not sample or execute an alternative action and does not change live PPO inputs.

For sustained-three, about 90.36% of honest local observations differ from the shadow, and mean action-distribution total variation is 0.1846. These measurements show signal visibility and sensitivity; they are not validated exposure labels or proof that each changed action is harmful.

After two trajectories diverge, comparing actions at the same world step is not an equal-state causal comparison.

7.6 Tools verification and reproducibility

The recorded 6 November full pinned Python 3.12 suite passed 172 tests, compared with the 162-test baseline before the diagnostic additions. Checks cover field conservation, local observations, shared reward, attack budgets, paired configurations, resume integrity, complete families, finite values, legal actions and leakage boundaries. Diagnostic regressions additionally verify shadow decay/capping, action-stream and RNG preservation, repeated exports, incomplete timeout state and selected-only decoy movement.

Evaluation preserved checkpoint bytes and parameters. Manifests record configurations, seeds, source fingerprints and completed/failed/incomplete state. Fresh output directories prevent silent reuse or overwriting of historical evidence. Compact episode-level CSVs, source hashes and manifests accompany the follow-up report; large checkpoints and step records remain local ignored artifacts [12]. The earlier raw corpus remains a recorded rather than freshly regenerated result.

The source review for this report checked all nineteen Python modules, fifteen test files and forty-four JSON configurations, plus decision records and saved result summaries. The old local training interpreter is unavailable on the current machine state, so the full 172-test result is dated evidence, not a fresh ESA rerun. Source parsing, configuration parsing and stored artifact checks are distinct from executing the training suite.

7.7 Response to review inputs

The panel concerns were the dataset-generation path and implementation complexity. We addressed them through staged clean calibration, frozen attack controls and a concrete corpus/CSV pipeline. Module tests and small end-to-end checks reduce the chance of mistaking a logging success for a research result. Shared faculty guidance also asked for a precise gap, new code and broader literature work. The critical comparison and new code are visible, while the wider survey and detector milestones remain open.

CHAPTER 8

CONCLUSION OF CAPSTONE PROJECT PHASE 1 AND FUTURE WORK

8.1 Phase 1 conclusion

We established a reproducible simulated retrieval task and a parameter-shared PPO pipeline with local observations. Candidate J supports a clean stochastic pheromone-benefit claim under the declared development comparison. We implemented bounded attack evaluation, preserved negative calibrations and built a validated development corpus and CSV export boundary.

The stronger sustained profile now demonstrates an average delivery reduction, but its case-to-case variation remains substantial. The local decoy route did not add mean injection damage. No trained detector, successful learned defense, final labelled dataset or held-out robustness result is claimed. These limitations guide the next experiments rather than being removed from the report.

8.2 Remaining work

The next scientific milestone is independent confirmation of the candidate attack under a predeclared development protocol. It should assess new development maps/action seeds and uncertainty without using final withheld conditions to tune the attack. Exposure targets must then account for residual injected pheromone and the effects of clipping and evaporation.

Complete maps and related episode families will be assigned to partitions before temporal windows. Detector selection and mitigation thresholds must use training/validation groups only. After those stages, we can collect learned-defense trajectories and compare undefended, no-pheromone, timeout and cautionary-pheromone baselines. Final tests must include unseen maps and an attack configuration withheld from development, together with clean cost, false alarms, delay, recovery and runtime.

Literature work continues toward the faculty's thirty-plus-paper expectation. Potential applications remain simulation motivations until the sensing and deployment assumptions are separately tested. Phase 1 has produced a working research harness and useful negative as well as positive development evidence; it has not completed the broader robustness study.

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APPENDIX A DEFINITIONS ACRONYMS AND ABBREVIATIONS

Term	Meaning in this project
Agent	One simulated worker choosing actions from local observations
Stigmergy	Indirect coordination through traces in the environment
PPO	Proximal Policy Optimization; the shared learned-policy method
MARL	Multi-agent reinforcement learning
Episode	One run until all food is delivered or the horizon is reached
World transition	One simultaneous action/update round for the whole team
Agent transition	One agent's record within a world transition
Seed	A reproducible control for training, map or action randomness
Corpus	An organised collection of trajectories and their provenance
Injection-disabled	Matched attack setup with false-food deposition switched off
Exposure target	A future validated label of locally observed injected signal
Provenance	Run, configuration, map, seed and group tracing information
Leakage	Information entering model inputs or test partitions improperly
Frozen policy	A checkpoint evaluated without updating its weights
Phase 2A and 2B	Internal work packages within academic Phase 1